

NOTE :

- text marked in grey are command to execute
- text marked in green are text to be modified in files

Prerequisites

System update

```
sudo apt-get upgrade
```

```
sudo apt-get update
```

1)first check whether java is install if yes then which version is installed

```
java -version
```

2)if not then run this command to install java

```
sudo apt-get install default-jre
```

3)after java install check for jps(Java Process Status) command working

```
jps
```

4)if not then run this command

```
sudo apt-get install openjdk-8-jdk
```

5)install ssh for remote login into data nodes

```
sudo apt-get install openssh-server
```

6)create a ssh public and private key

```
ssh-keygen
```

7)add the pub to key the authorised key file for password less access

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

8)try whether it works or not

```
ssh localhost
```

9)to get out use exit cmd

```
exit
```

10)now Untar the hadoop.tar.gz file

```
tar -xvzf < path to hadoop  
downloaded file > hadoop-2.9.1.tar.gz
```

11)create a soft link for hadoop folder (optional)

```
ln -s hadoop-<Version of hadoop> hadoop
```

12)locate your java path

```
readlink -f $(which java)
```

```
# output will be /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java
```

```
select only /usr/lib/jvm/java-8-openjdk-amd64
```

13) now open bashrc file and add Java and Hadoop Path

```
gedit ~/.bashrc
```

#inside this file at the last, add these line in green colour

```
export JAVA_HOME=< java path as seen in above point 12 >
export HADOOP_HOME=<path where hadoop downloaded file was extracted>
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

example:-

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/home/cdac/Desktop/hadoop-3.2.0
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

#Save the file and source the file using

```
source ~/.bashrc
```

14)Now try hadoop version

```
hadoop version
```

output will the version of hadoop installed

15)Now again open the ~/.bashrc file and add the following lines

#NOTE ONLY ADD THOSE LINES WHICH ARE NOT THERE THE FINAL FILE SHOULD LOOK LIKE THE FOLLOWING (as marked in green)

```
gedit ~/.bashrc
```

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/home/cdac/Desktop/hadoop-2.7.3
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export HADOOP_YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
```

16)Save the file and source the file using

```
source ~/.bashrc
```

17)NOW go to Hadoop folder

```
cd $HADOOP_HOME/etc/hadoop
```

17.1.)Open hadoop-env.sh file

```
gedit hadoop-env.sh
```

#at the last line of file add

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

#save the file

17.2) In the same directory i.e

\$HADOOP_HOME/etc/hadoop cp mapred-site.xml.template mapred-site.xml

open this file mapred-site.xml

gedit mapred-site.xml

#add following property line in between configuration line (marked in green)

```
<configuration>
```

```
<property>
```

```
<name>mapreduce.framework.name</name>
```

```
<value>yarn</value>
```

```
</property>
```

```
</configuration>
```

#save the file

17.3.) similarly open core-site.xml

#add following property line in between configuration line (marked in green)

```
<configuration>
```

```
<property>
```

```
<name>fs.defaultFS</name>
```

```
<value>hdfs://localhost:9000</value>
```

```
</property>
```

```
</configuration>
```

#save the file

17.4) now create tow dir in your system where you want to store your data of namenode and datanode

1. mkdir -p /home/cdac/Desktop/hdfs/namenode
2. mkdir -p /home/cdac/Desktop/hdfs/datanode

gedit hdfs-site.xml

#add following property line in between configuration line (marked in green)

```
<configuration>
```

```
<property>
```

```
<name>dfs.replication</name>
```

```
<value>1</value>
```

```
</property>
```

```
<property>
```

```
<name>dfs.namenode.name.dir</name>
```

```
<value>/home/cdac/Desktop/hdfs/namenode</value>
```

```
</property>
```

```
<property>
```

```
<name>dfs.datanode.data.dir</name>
```

```
<value>/home/cdac/Desktop/hdfs/datanode</value>
```

```
</property>
```

```
</configuration>
```

#save the file

17.5.) open this file yarn-site.xml

gedit yarn-site.xml

#add following property line in between configuration line (marked in green)

```
<configuration>
```

```
<property>
```

```
<name>yarn.nodemanager.aux-services</name>
```

```
<value>mapreduce_shuffle</value>
```

```
</property>
```

```
</configuration>
```

#save the file

18) Now format your

namenode #NOTE this

should be done only once

hadoop namenode -format

#output of this cmd should be succesfull like this(marked in green)

18/06/23 12:09:36 INFO common.Storage: Storage directory

/home/rohit/Desktop/hdfs/namenode has been successfully formatted.

18/06/23 12:09:36 INFO namenode.FSImageFormatProtobuf: Saving image file

/home/rohit/Desktop/hdfs/namenode/current/fsimage.ckpt_00000000000000000000

using no compression

18/06/23 12:09:36 INFO namenode.FSImageFormatProtobuf: Image file

/home/rohit/Desktop/hdfs/namenode/current/fsimage.ckpt_00000000000000000000 of size
322 bytes saved in 0 seconds .

19) now start the hadoop by using

start-dfs.sh

start-yarn.sh

#after that check jps

jps

#it should display the following processes running

22672 ResourceManager

23088 Jps

22499 SecondaryNameNode

22804 NodeManager

22300 DataNode

22143 NameNode

Your hadoop is now up and runing you could check its ui at url

<http://localhost:50070>

20) similarly to stop
hadoop use stop-yarn.sh

Open the below links:

1.) For NameNode:
<http://localhost:50070>

2.) For Resource Manager:
<http://localhost:8088>

3.) For Secondary NameNode:
<http://localhost:50090>

4.) For DataNode:
<http://localhost:50075>

To Stop:

1.) stop-dfs.sh

2.) stop-yarn.sh